

Homework 1: Scalar flows, modeling, and critical slowing down

MATH 515 — Fall 2026

Due Friday, September 11, 2026

Assignment convention. Each numbered problem reproduces the full assigned statement from Steven H. Strogatz, *Nonlinear Dynamics and Chaos*, third edition (the specific section/problem number is given parenthetically), with some mild editing for clarity. Any additional requirements are labeled **Course-specific instructions** and are part of the assignment.

Problem 1: Implicit fixed points

Textbook problem (Strogatz, third edition, Exercise 2.2.7):

Analyze the equation graphically. Sketch the vector field on the real line, find all the fixed points, classify their stability, and sketch the graph of $x(t)$ for different initial conditions. Then try for a few minutes to obtain the analytical solution for $x(t)$; if you get stuck, do not try for too long, since it may be impossible to solve the equation in closed form:

$$\dot{x} = e^x - \cos x.$$

Hint: Sketch the graphs of e^x and $\cos x$ on the same axes, and look for intersections. You will not be able to find the fixed points explicitly, but you can still find the qualitative behavior.

Course-specific instructions. Complete the exercise. A graphical assertion by itself is not enough: justify the number and location of the fixed points, as well as their stability, using appropriate analytical properties of the functions involved.

Problem 2: Constructing flows from prescribed fixed points

Textbook problem (Strogatz, third edition, Exercise 2.2.10, “Fixed points”):

For each of (a)–(e), find an equation $\dot{x} = f(x)$ with the stated properties, or if there are no examples, explain why not. (In all cases, assume that $f(x)$ is a smooth function.)

- (a) Every real number is a fixed point.
- (b) Every integer is a fixed point, and there are no others.
- (c) There are precisely three fixed points, and all of them are stable.
- (d) There are no fixed points.
- (e) There are precisely 100 fixed points.

Course-specific instructions. Complete all parts. In each realizable case, verify that your proposed right-hand side is smooth and has exactly the requested fixed points. In each impossible case, give a proof.

Problem 3: Terminal velocity [Challenge]**Textbook problem** (Strogatz, third edition, Exercise 2.2.13, “Terminal velocity”):The velocity $v(t)$ of a skydiver falling to the ground is governed by

$$m\dot{v} = mg - kv^2,$$

where m is the mass of the skydiver, g is the acceleration due to gravity, and $k > 0$ is a constant related to the amount of air resistance.

- Obtain the analytical solution for $v(t)$, assuming that $v(0) = 0$.
- Find the limit of $v(t)$ as $t \rightarrow \infty$. This limiting velocity is called the terminal velocity. (Beware of bad jokes about the word *terminal* and parachutes that fail to open.)
- Give a graphical analysis of this problem, and thereby re-derive a formula for the terminal velocity.
- An experimental study (Carlson et al. 1942) confirmed that the equation $m\dot{v} = mg - kv^2$ gives a good quantitative fit to data on human skydivers. Six men were dropped from altitudes varying from 10,600 feet to 31,400 feet to a terminal altitude of 2,100 feet, at which they opened their parachutes. The long free fall from 31,400 to 2,100 feet took 116 seconds. The average weight of the men and their equipment was 261.2 pounds. In these units, $g = 32.2 \text{ ft/sec}^2$. Compute the average velocity V_{avg} for the long free fall.
- Using the data given here, estimate the terminal velocity and the value of the drag constant k .

Hints: First you need to find an exact formula for $s(t)$, the distance fallen, where $s(0) = 0$, $\dot{s} = v$, and $v(t)$ is known from part (a). You should get $s(t) = \frac{V^2}{g} \ln\left(\cosh \frac{gt}{V}\right)$, where V is the terminal velocity. Then solve for V graphically or numerically, using $s = 29,300$, $t = 116$, and $g = 32.2$. A slicker way to estimate V is to suppose $V \approx V_{\text{avg}}$ for the long free fall, as a rough first approximation. Then show that $gt/V \approx 15$. Since $gt/V \gg 1$, we may use the approximation $\ln(\cosh x) \approx x - \ln 2$ for $x \gg 1$. Derive this approximation and then use it to obtain an analytical estimate of V . Then k follows from part (b).

This analysis is from Davis (1962).

Course-specific instructions. Complete all parts. Carry units through the data analysis in parts (d)–(e). Give both a numerical estimate obtained from the exact distance equation and the large-argument approximation requested in the exercise, and compare the two estimates. For the exact equation, you may use bisection—repeatedly halving an interval on which the residual has opposite signs at its endpoints—or another reproducible scalar root solver. Here define

$$R(V) = \frac{V^2}{g} \log\left(\cosh \frac{gt}{V}\right) - s,$$

where s, t, g are the given data. A *bracket* is an interval $[V_-, V_+]$ with $R(V_-)R(V_+) < 0$. State the initial bracket and a stopping tolerance, meaning either a maximum final bracket width or a maximum value of $|R(V)|$. **Computational submission.** Include code that would allow another person to reproduce the calculation or plot. Clearly label every reported numerical value and plot, and explain what the output shows.

Problem 4: Language death

Textbook problem (Strogatz, third edition, Exercise 2.3.6, “Language death”):

Thousands of the world’s languages are vanishing at an alarming rate, with 90 percent of them being expected to disappear by the end of this century. Abrams and Strogatz (2003) proposed the following model of language competition, and compared it to historical data on the decline of Welsh, Scottish Gaelic, Quechua (the most common surviving indigenous language in the Americas), and other endangered languages.

Let X and Y denote two languages competing for speakers in a given society. The proportion of the population speaking X evolves according to

$$\dot{x} = (1 - x)P_{YX} - xP_{XY},$$

where $0 \leq x \leq 1$ is the current fraction of the population speaking X , $1 - x$ is the complementary fraction speaking Y , and P_{YX} is the rate at which individuals switch from Y to X . This deliberately idealized model assumes that the population is well mixed (meaning that it lacks all spatial and social structure) and that all speakers are monolingual.

Next, the model posits that the attractiveness of a language increases with both its number of speakers and its perceived status, as quantified by a parameter $0 \leq s \leq 1$ that reflects the social or economic opportunities afforded to its speakers. Specifically, assume that $P_{YX} = sx^a$ and, by symmetry, $P_{XY} = (1 - s)(1 - x)^a$, where the exponent $a > 1$ is an adjustable parameter. Then the model becomes

$$\dot{x} = s(1 - x)x^a - (1 - s)x(1 - x)^a.$$

- Show that this equation for $\dot{x}$ has three fixed points.
- Show that for all $a > 1$, the fixed points at $x = 0$ and $x = 1$ are both stable.
- Show that the third fixed point, $0 < x^* < 1$, is unstable.

This model therefore predicts that two languages cannot coexist stably—one will eventually drive the other to extinction. For a review of generalizations of the model that allow for bilingualism, social structure, etc., see Castellano et al. (2009).

Course-specific instructions. Complete all parts with the assumption $a > 1$ and $0 < s < 1$ (i.e. you may ignore the cases where $s = 0$ and $s = 1$). Your stability analysis of the interior equilibrium must apply for arbitrary admissible a and s , not only for a numerical example.

Problem 5: Critical slowing down

Textbook problem (Strogatz, third edition, Exercise 2.4.9, “Critical slowing down”):

In statistical mechanics, the phenomenon of “critical slowing down” is a signature of a second-order phase transition. At the transition, the system relaxes to equilibrium much more slowly than usual. Here is a mathematical version of the effect:

- Obtain the analytical solution to $\dot{x} = -x^3$ for an arbitrary initial condition. Show that $x(t) \rightarrow 0$ as $t \rightarrow \infty$, but that the decay is not exponential. (You should find that the decay is a much slower algebraic function of t .)

- (b) To get some intuition about the slowness of the decay, make a numerically accurate plot of the solution for the initial condition $x_0 = 10$, for $0 \leq t \leq 10$. Then, on the same graph, plot the solution to $\dot{x} = -x$ for the same initial condition.

Course-specific instructions. Complete both parts. Make the comparison plot reproducible and label the algebraic and exponential decays clearly. **Computational submission.** Include code that would allow another person to reproduce the calculation or plot. Clearly label every reported numerical value and plot, and explain what the output shows.